

SAMANTHA LAVRINC

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SUMMARY

Software Engineer with 3+ years of experience developing clinical NLP pipelines and backend services in healthcare technology. Skilled in Java, SQL, and cloud-based CI/CD, with expertise in extracting structured insights from unstructured medical text. Adept at bridging technical and clinical teams to translate medical requirements into scalable engineering solutions.

WORK EXPERIENCE

Software Engineer - NLP & Clinical Informatics, Realize Intelligence **Oct 2022 - Present**

- Maintained and enhanced a multi-layer NLP pipeline leveraging Apache cTAKES, Java, and PostgreSQL to process unstructured clinical records into FHIR-compliant JSON.
- Debugged and optimized existing finite state machines, regex-driven negation engines, and terminology lookup systems to ensure accurate detection of dates, measurements, and clinically relevant concepts.
- Designed and implemented extensible NLP components to enhance rule-based pipelines with advanced inference, structured data handling, and cross-entity coordination.
- Extended regex-based negation and pattern-matching logic to improve detection of clinically relevant concepts.
- Implemented comprehensive JUnit test coverage for all new and modified engines, ensuring pipeline reliability and preventing regressions.
- Collaborated weekly with the Chief Medical Officer to translate clinical trial protocols into technical requirements, building value sets, rules, and tagging configurations for oncology trial matching.
- Responsible for maintaining and refining system updates and clinical logic following senior departures, helping ensure consistent and accurate processing across thousands of patient records.
- Bridged communication between clinical stakeholders and engineering teams by researching oncology and genomics concepts, enabling informed technical decisions and fostering trust across disciplines.

Software Engineer Intern – UPMC Information Services **Jun 2021 - Aug 2021**

- Redesigned Azure CI/CD pipelines to support separate production and review workflows, converting Classic pipelines to YAML for maintainability and alignment with Scrum release processes.
- Gained practical experience with Azure DevOps, C#, YAML, and healthcare security standards (HIPAA).

EDUCATION

Bachelor of Science in Computer Science **Sep 2020 - Aug 2022**

University of Pittsburgh - School of Computing and Information

- Dean's List - 2021

Associate of Science in Computer Information Systems **Aug 2019 - Aug 2020**

Community College of Allegheny County

- Dean's List 2019 - 2020
- Phi Theta Kappa Honors Society

TECHNICAL SKILLS

- Programming: Java (proficient), Python, C#, C++ (familiar), HTML/CSS/JavaScript (familiar)
- Databases & ETL: PostgreSQL, SQL, ETL/Data Integration
- NLP & Healthcare: cTAKES, Regex, FHIR/HL7, SNOMED, LOINC, RxNorm, ICD-O/ICD-10
- CI/CD & Testing: JUnit, Groovy, Cucumber, Azure DevOps, Docker, YAML pipelines
- Cloud & Microservices: AWS
- Collaboration Tools: Jira, Confluence